Quantifying Network Similarity using Graph Cumulants

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TL;DR: [1] Graph cumulants perform better [2] and are more intuitive than the typical subgraph statistics

Using only moments is **awkward** [3]

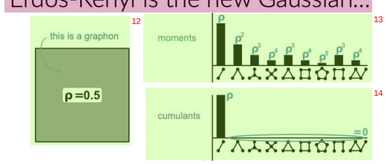
"The length of a human averages 1.7 meters, [4]
and their **average squared length** is 2.9 square meters" [5]

Using cumulants is **easier** to understand [6]

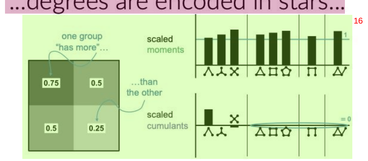
"The length of a human has: a **variance** of 0.01 meters, [7]
a **standard deviation** of 0.1 meters, [8]
a **relative fluctuation** of 6%" [9]

Graph Cumulants: the Better Subgraph Statistics [10]

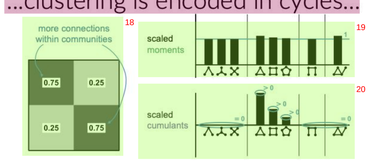
Erdős-Rényi is the new Gaussian... [11]



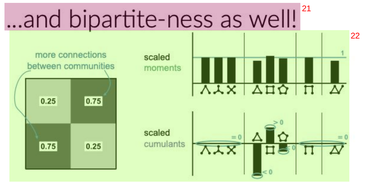
...degrees are encoded in stars... [15]



...clustering is encoded in cycles... [17]

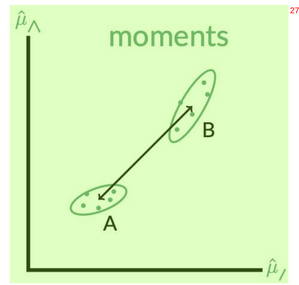


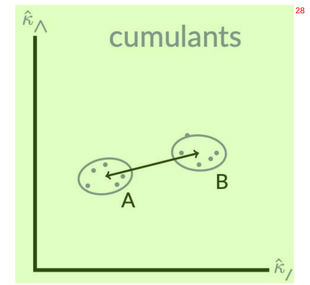
...and bipartite-ness as well! [21]



But are graph cumulants better for testing? [23] [24]
(than subgraph densities) [25]

Apples-to-Apples: A Two-Sample Test [26]





$$\hat{d}^2_\kappa(\mathsf{A},\mathsf{B}) = \left(\hat{\boldsymbol{\kappa}}(\mathsf{A}) - \hat{\boldsymbol{\kappa}}(\mathsf{B})\right)^\top \left(\hat{\Sigma}_\kappa(\mathsf{A}) + \hat{\Sigma}_\kappa(\mathsf{B})\right)^{-1} \left(\hat{\boldsymbol{\kappa}}(\mathsf{A}) - \hat{\boldsymbol{\kappa}}(\mathsf{B})\right)$$ [29]

$$\underbrace{\hat{d}^2_\mu(\mathsf{A},\mathsf{B})}_{\text{"}Z^2\text{-score"}} = \underbrace{\left(\hat{\boldsymbol{\mu}}(\mathsf{A}) - \hat{\boldsymbol{\mu}}(\mathsf{B})\right)}_{\text{difference}}^\top \underbrace{\left(\hat{\Sigma}_\mu(\mathsf{A}) + \hat{\Sigma}_\mu(\mathsf{B})\right)^{-1}}_{\text{metric}} \underbrace{\left(\hat{\boldsymbol{\mu}}(\mathsf{A}) - \hat{\boldsymbol{\mu}}(\mathsf{B})\right)}_{\text{difference}}$$ [30]

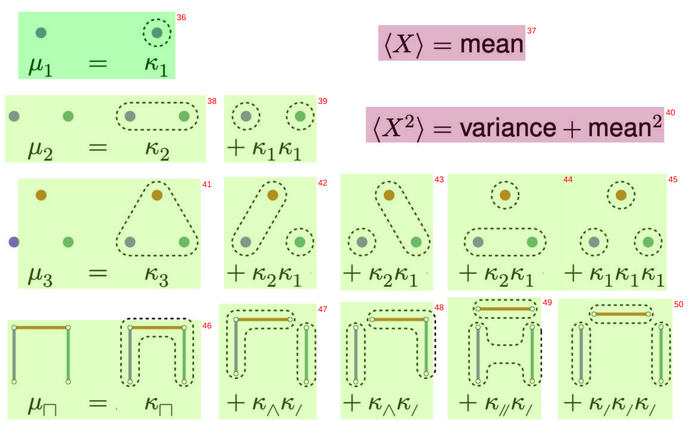
When estimating the covariance... [31]

$$\mathrm{Cov}(\hat{\mu}_g, \hat{\mu}_{g'}) = \underbrace{\langle \hat{\mu}_g\ \hat{\mu}_{g'} \rangle}_{\text{"hard"}} - \underbrace{\langle \hat{\mu}_g \rangle \langle \hat{\mu}_{g'} \rangle}_{\text{"easy"}}$$ [32]

...the "hard" part uses a combinatorial disjoint union rule [33]

$$c_\wedge c_/ = 4c_\wedge + 2c_\Delta + 2c_{\curlywedge} + 4c_\sqcap + c_{/\!/\wedge}$$ [34]

Combinatorial Construction of Cumulants [35]

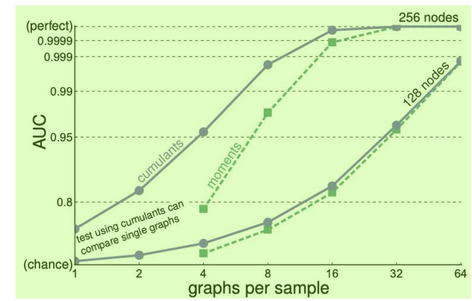


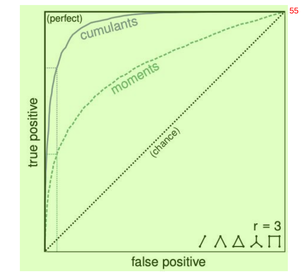
$\langle X \rangle = \text{mean}$ [37]

$\langle X^2 \rangle = \text{variance} + \text{mean}^2$ [40]

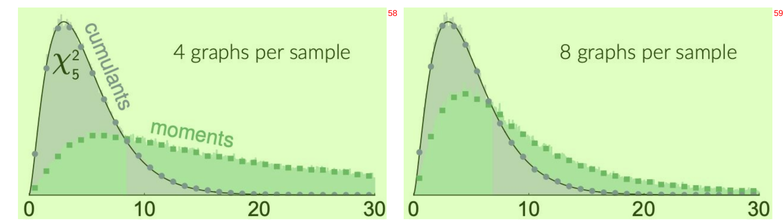
Graph Cumulants Clearly Conquer [51]

Graph cumulants outperform subgraph densities in general... [52]
...and graph cumulants also work for single graph samples! [53]



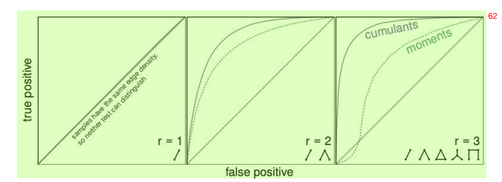


Why do graph cumulants perform better? [56]
...because their **fluctuations** look more $\mathcal{N}$ormal! [57]



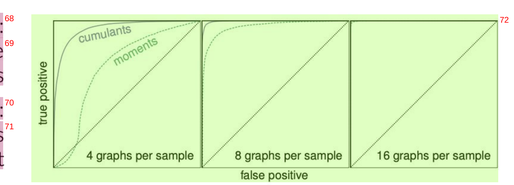
"Z^2-score" for graphs sampled from the same distribution [60]

Graph Cumulants in the (semi-)Wild! [61]



Varying: [63]
Number of subgraphs [64]
used by both tests [65]

Comparing: [66]
Genetic interaction networks [67]
of Arabidopsis and Mouse

Varying: [68]
Number of graphs per sample [69]
used by both tests

Comparing: [70]
Genetic interaction networks [71]
of Human and Rat